PHYS4055 Week 5 Tutorial Questions

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Tutorial Question 1: A Pure-Loss Channel and its Kraus Representation

A single-rail photonic qubit is prepared in the state

$$|\psi\rangle_S = \alpha|0\rangle_S + \beta|1\rangle_S, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{1}$$

A pure-loss channel with transmissivity $0 \le \eta \le 1$ is modelled by coupling the system to an environment initially in the vacuum state $|0\rangle_E$. The joint system–environment evolution is

$$|0\rangle_S|0\rangle_E \longrightarrow |0\rangle_S|0\rangle_E, \tag{2}$$

$$|1\rangle_S|0\rangle_E \longrightarrow \sqrt{\eta}\,|1\rangle_S|0\rangle_E + \sqrt{1-\eta}\,|0\rangle_S|1\rangle_E. \tag{3}$$

1. Find the joint system–environment state after the interaction.

New tool: Kraus representation

In this problem, the environment is not accessed or measured. The output state of the system is obtained by tracing over the environment:

$$\hat{\rho}'_S = \mathrm{Tr}_E\left[\hat{U}_\eta\,(\hat{\rho}_S \otimes |0\rangle_E\langle 0|)\,\hat{U}_\eta^\dagger\right]. \tag{4}$$

To evaluate the partial trace, choose an orthonormal basis $\{|j\rangle_E\}$ for the environment and insert

$$\hat{I}_E = \sum_j |j\rangle_E\langle j|. \tag{5}$$

The reduced system state can then be written as

$$\hat{\rho}'_S = \sum_j {}_E\langle j|\hat{U}_\eta\,(\hat{\rho}_S \otimes |0\rangle_E\langle 0|)\,\hat{U}_\eta^\dagger|j\rangle_E \tag{6}$$

$$= \sum_j \hat{K}_j\hat{\rho}_S\hat{K}_j^\dagger, \tag{7}$$

where

$$\hat{K}_j = {}_E\langle j|\hat{U}_\eta|0\rangle_E. \tag{8}$$

The operators $\hat{K}_j$ are called *Kraus operators*. The index j labels an environment basis component used when performing the partial trace; it does not imply that the environment is measured.

Each Kraus operator gives an unnormalised contribution

$$\widetilde{\rho}_{S,j} = \hat{K}_j\hat{\rho}_S\hat{K}_j^\dagger. \tag{9}$$

The trace

$$w_j = \mathrm{Tr}\,(\widetilde{\rho}_{S,j}) \tag{10}$$

is the weight of this contribution in the reduced system state.

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Since the environment is not observed, the individual contributions are not selected or normalised separately. The physical output state is their sum:

$$\hat{\rho}'_S = \sum_j \widetilde{\rho}_{S,j}. \tag{11}$$

The complete channel preserves the trace provided that

$$\sum_j \hat{K}_j^\dagger \hat{K}_j = \hat{I}_S. \tag{12}$$

2. Use the beam-splitter transformations and the definition of the Kraus operators to show that, within the single-rail subspace,

$$\hat{K}_0 = |0\rangle\langle 0| + \sqrt{\eta}\,|1\rangle\langle 1|, \tag{13}$$
$$\hat{K}_1 = \sqrt{1-\eta}\,|0\rangle\langle 1|. \tag{14}$$

   Verify explicitly that

$$\hat{K}_0^\dagger \hat{K}_0 + \hat{K}_1^\dagger \hat{K}_1 = \hat{I}_S. \tag{15}$$

   Explain why this condition ensures that the complete channel preserves the trace.

3. For each Kraus operator, calculate the unnormalised state vector

$$|\phi_j\rangle_S = \hat{K}_j |\psi\rangle_S \tag{16}$$

   and the corresponding contribution

$$\widetilde{\rho}_{S,j} = |\phi_j\rangle\langle\phi_j|. \tag{17}$$

   Calculate the weights

$$w_j = \langle\phi_j|\phi_j\rangle \tag{18}$$

   and verify that

$$w_0 + w_1 = 1. \tag{19}$$

   Identify which Kraus contribution represents the no-loss component and which represents the photon-loss component.

   Explain why the individual contributions $\widetilde{\rho}_{S,j}$ do not generally have unit trace, and why the physical reduced system state must include both contributions.

4. Using

$$\hat{\rho}_S = |\psi\rangle\langle\psi|, \tag{20}$$

   calculate the reduced system state

$$\hat{\rho}'_S = \hat{K}_0 \hat{\rho}_S \hat{K}_0^\dagger + \hat{K}_1 \hat{\rho}_S \hat{K}_1^\dagger. \tag{21}$$

   Write the result explicitly in the basis $\{|0\rangle, |1\rangle\}$.

   Explain why the one-photon population is multiplied by η, while the coherence between $|0\rangle$ and $|1\rangle$ is multiplied by $\sqrt{\eta}$.

   Explain why the reduced system state can be mixed even though the initial system state and the joint system–environment output state are both pure. In particular, discuss the role of the orthogonal environment states associated with the loss and no-loss components.

5. Calculate the purity of the reduced output state and show that

$$\mathrm{Tr}\left[(\hat{\rho}'_S)^2\right] = 1 - 2\eta(1-\eta)|\beta|^4. \tag{22}$$

For $0 < \eta < 1$, determine which input state remains pure after the channel. Explain how the reduction in system purity is related to correlations between the system and environment.

6. Now consider the particular input state

$$|\psi\rangle_S = |1\rangle_S. \tag{23}$$

Calculate the output probabilities for detecting zero photons and one photon. Hence calculate

$$\langle\hat{n}\rangle, \qquad \langle\hat{n}^2\rangle, \tag{24}$$

where

$$\hat{n} = |1\rangle\langle 1| \tag{25}$$

within the single-rail subspace.

Show that the output photon-number variance is

$$\mathrm{Var}(\hat{n}) = \langle\hat{n}^2\rangle - \langle\hat{n}\rangle^2 = \eta(1-\eta). \tag{26}$$

Determine the value of η for which the photon-number uncertainty is greatest, and explain why the variance vanishes at both $\eta = 0$ and $\eta = 1$.